Check for updates

Original Article

autism

Autism
1–18
© The Author(s) 2021
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/1362361320984899
journals.sagepub.com/home/aut
SAGE

Can pictorial narration offer a solution to teacher training on the effective inclusion of students with autism spectrum disorder in low-resource settings? Investigation on knowledge and stigma change

Hui Min Low[1], Tze Peng Wong[2], Lay Wah Lee[1], Somchay Makesavanh[3], Bountheing Vongsouangtham[3], Vikate Phannalath[3], Aznan Che Ahmad[1] and Ann Sien Sut Lee[1]

Abstract

The main aim of this article is to explore whether pictorial narration could offer a solution to teacher training on the effective inclusion of students with autism spectrum disorder in a low-resource context in the Lao People's Democratic Republic. For this purpose, pre- and post-training knowledge data were collected from 87 Laotian teachers who participated in teacher training using a pictorial narrative e-module called The Story of Khamdy™. The teacher training included a half-day orientation workshop, followed by 4 weeks of online learning. Statistically significant improvement in autism spectrum disorder knowledge and reduction in autism spectrum disorder stigma was recorded for the teachers after completing the teacher training. The findings on social validity pointed to high acceptance toward the pictorial narrative method and the content designed for the purpose of this training. The teachers' acceptance of the training method was found to have positive effects on their knowledge changes in the diagnosis and etiology aspects of autism spectrum disorder knowledge. The findings supported the use of a pictorial narration approach in imparting knowledge about inclusive education and autism spectrum disorder to teachers in a least developed country. The findings offered insights to explain the knowledge change and stigma reduction recorded.

Lay abstract

In this study, we explored whether pictorial narration could offer a solution to teacher training on effective inclusion of students with autism spectrum disorder in the Lao People's Democratic Republic. For this purpose, pre- and post-training knowledge data were collected from 87 Laotian teachers who participated in teacher training using a pictorial narrative e-module called The Story of Khamdy™. The teachers' knowledge test results and feedback were analyzed. The findings indicated that teachers' acceptance toward the training method had positive effects on their knowledge changes and supported the use of a pictorial narration approach in imparting knowledge about inclusive education and autism spectrum disorder to teachers in a least developed country.

Keywords

autism spectrum disorders, inclusion, knowledge, Lao People's Democratic Republic, low- and middle-income countries, stigma, teachers

Introduction

In implementing inclusive education for young children with autism spectrum disorder (hereafter, ASD), teachers of mainstream classes empowered to apply inclusive practices are more likely to ensure successful learning for these children (Shoulders & Krei, 2016). Despite an increased awareness of ASD in many parts of the world, teachers

[1]Universiti Sains Malaysia, Malaysia
[2]University of Nottingham Malaysia Campus, Malaysia
[3]Souphanouvong University, Lao People's Democratic Republic

Corresponding author:
Hui Min Low, School of Educational Studies, Universiti Sains Malaysia, Pulau Pinang 11800, Malaysia.
Email: lowhm@usm.my

from under-resourced contexts continue to report a lack of knowledge about ASD and the skills to include and accommodate the learning needs of students with ASD in their existing schools (Ravet, 2018). This knowledge gap is undeniably one of the major barriers in the provision of inclusive education to young children with ASD, resulting in negative consequences, such as limited learning opportunities, ineffective learning, and negative academic outcomes (Meindl et al., 2020). The knowledge gap is even more pertinent in developing and least developed countries (Low et al., 2019; Ritter et al., 2019). While these countries could utilize resources established by other countries (Parsons & Kovshoff, 2019), cultural and language barriers could possibly reduce the suitability of those resources in helping teachers plan and execute effective lessons for children with ASD in their classrooms.

Pictorial communication

Grounded within a low-resourced context, in this article, we explored whether pictorial narration could offer a solution to improve teachers' knowledge and skills to include children/students with ASD in their classrooms. To the best of our knowledge, pictorial narration as a systematic knowledge transfer method for the specific purpose of teacher training in the area of ASD has yet to be reported. The idea of incorporating pictorial narration in ASD teacher training was inspired by the commonplace of utilizing pictorial communication in the field of medical and health sciences to improve health literacy among the general public (Brewer et al., 2016) and the minority community (Thunberg et al., 2019). Broadly, health literacy refers to the ability to read and interpret medical-related information (Sørensen & Pleasant, 2017). It encompasses knowledge, motivation, and competence to access, understand, appraise, and apply information to make decisions related to disease prevention and health promotion (Quaglio et al., 2017). The devastating impact of low health literacy is especially prominent in lower income communities, which leads to inadequate access to health care, reduced medical compliance, and over-reliance on non-scientific advice or solutions (Colledge et al., 2008; Edward et al., 2018; Swartz et al., 2018). Upon realizing that many people find it hard to understand and act on medical information, health practitioners disseminated information in pictorial forms to aid understanding of health and medical care among people of all levels (Colledge et al., 2008; Edward et al., 2018; Weaver et al., 2019). For example, Koops van't Jagt et al. (2016) developed seven photo stories to aid doctor–patient communication involving older adults with limited health literacy in the Netherlands, and this method was well-received by their senior-age participants.

There are a few reasons why a narrative form of pictorial communication is effective in disseminating health information to specific populations with low health literacy. As outlined by Koops van't Jagt et al. (2016), this form of knowledge dissemination is educational yet entertaining. More importantly, it provides a familiar form of interaction, that is, story-telling, which is more likely to increase personal involvement and motivation to accept and act on the message received. Given that, it helps the readers or viewers to relate themselves to the characters in the story and enhance their self-efficacy to perform the suggested behaviors (Koops van't Jagt et al., 2016). So far, this narrative-based pictorial communication technique has recorded success in disseminating information and knowledge related to diabetes information, depression, smoking effects, and vaccination to people with low level of health literacy, especially when it presents cultural familiarity to the audience (Kaur, 2019).

The application of pictorial communication in health and medical sciences parallels the aim of transmedia communication, namely the cognitive transmission of information between (at least) two minds, that is, the producer's mind and the perceiver's mind, through the aid of a media product (Elleström, 2019). According to Elleström (2019), in transmedia communication, the perceiver configures his or her mental understanding of concepts (namely the virtual sphere) to match, as much as possible, concepts in the producer's mind. The use of pictorial communication has been further explained by dual coding theory (Paivio, 1991), in which people were found to remember and engage more with the information provided through both visual and verbal modes, especially when texts and graphics are mapped in a meaningful way (Bresciani et al., 2018). Therefore, in the health and medical industry, pictorial communication through informative posters and pictograms in brochures is especially effective in circumstances where medical service providers and patients do not speak the same language (Clawson et al., 2012). In such circumstances, pictures act as intermediate entities to bridge information transfer between the medical service provider and the service recipients (Geana et al., 2012). Further, pictorial communication was also found to enhance satisfaction and improve information recall among the service recipients (Meppelink et al., 2016; Muturi, 2016).

Pictorial narration in teacher training of ASD

The strong theoretical and practical basis of implementing pictorial communication for the purpose of knowledge transfer in the medical industry provides the foundation for developing a teacher training program on ASD using pictorial communication. In fact, utilizing pictorial communication to disseminate ASD-related information is not new. There is an abundance of pictorial infographics, brochures, posters, and videos about ASD-related characteristics and management of ASD, either in the form of print resources or on various online platforms. However, in countries with low resources, existing pictorial resources from other cultures and

contexts may not fully suit their needs. The capacity of these resources in the effective transmission of new knowledge to certain communities may be limited. In addition, existing pictorial resources, mainly in the formats of infographics or posters, present information of a narrow scope or topic that has been highly summarized or simplified. For the purpose of teacher training in the area of ASD, these conventional pictorial resources are, therefore, not graphically sufficient to present the complexity and heterogeneity of ASD in all domains of development, especially when the teacher training also requires the transmission of knowledge in applying inclusive practices according to the individual needs of the child with ASD, as well as all other children in the classroom. Capitalizing on the advantages of pictorial narration in enhancing learning via a mode of engagement that is culturally familiar for the purpose of teacher training, we propose the use of an elaborated pictorial narration that systematically and comprehensively presents all essential topics related to ASD as well as inclusive practices that are contextualized to the existing beliefs and resources available in the community.

A further review of the literature on health communication in low-resource contexts points toward the use of "narrative" or "story-telling" as a solution to enhance meaning transmission through pictorial resources. Muturi (2016) argued that using "narratives" or "story-telling" is a meaningful and effective way to engage a local community with concepts and practices that are foreign to them. In a study that explores the use of communication strategies to prevent alcohol abuse in rural central Kenya, Muturi (2016) documented the potential of using narratives in health communication. Traditionally, in low-resource contexts like central Kenya, the older generation educated the younger generation by telling them stories about the past as a way to pass on wisdom (Muturi, 2016). As elaborated by a local woman in her study, "If you can make a cinema to tell a story of someone from the time they were healthy and follow up with them until they die, that might speak to a lot of people" (p. 314). The excerpt highlights two important notions about pictorial narration: (a) the notion of "a picture speaks a thousand words" and (b) the notion of transmission of culture-foreign information through a culturally acceptable mode, in this case, a narrative or story-telling.

In the context of our study, a narrative refers to the communication of real or fictive events and situations in a time sequence (Prince, 2012), which is constituted by a set of agents and events linked in a cause–effect fashion (Elleström, 2019). These narrative representations provided the basis for creating a Khamdy, a fictional child with ASD, for the purpose of teacher training in effective inclusion of students with ASD in the Lao People's Democratic Republic (Lao PDR), a country identified by the United Nations as one of the world's least developed countries (LDCs).

The teacher training project was supported by a charitable organization and involved cross-country collaboration between a group of researchers and practitioners in the area of special and inclusive education and ASD from Malaysia and local teacher trainers from the Lao PDR, which are both southeast Asian countries. One major challenge in the implementation of this project was the introduction of ASD, a concept that was considered foreign to the communities in the Lao PDR and a word that is not used in their language. To the best of this project researcher's knowledge, there were no local ASD-related resources and no existing vocabulary repertoire that the researchers could utilize for this intended teacher training on ASD. After identifying the potential of utilizing a pictorial-enhanced story-telling method to deliver novel technical contents to teachers in the Lao PDR, an e-module that encompassed a total of 20 short stories called The Story of Khamdy™ was jointly developed by the group of international researchers and local teacher trainers (Figure 1).

The short stories in the e-module were created through co-construction of knowledge and participatory decision-making during the process of module development (Abma et al., 2017), involving the international team members who have expertise in the area of ASD and inclusive practices, and local team members who have expertise in teacher training at the Lao PDR. The e-module was delivered via colorful pictures with an easy-to-understand script packaged in the MP4 format. Oral narration and visual animations were added to enhance the clarity of the content, the entertainment value of these videos, and engagement between the teachers and content. Each story is linked in a cause–effect fashion, featuring a Laotian boy with ASD called Khamdy, his family, teachers, and friends in school, encountering different challenges associated with ASD symptoms and how they proactively seek and solve the problems based on evidence-based solutions (Figure 2).

ASD knowledge and stigma

Although the effectiveness of pictorial narratives has been demonstrated for improving health literacy (Koops van't Jagt et al., 2016; Muturi, 2016), its effectiveness as a method for teacher training on the effective inclusion of students with ASD has yet to be explored. Teacher training is the catalyst of knowledge change and stigma reduction. Globally, ASD has been understood from different ASD knowledge dimensions, including knowledge about diagnosis, knowledge about etiology, and knowledge about treatment (Harrison, Slane et al., 2017). Knowledge about diagnosis (Harrison, Bradshaw et al., 2017) refers to the understanding of ASD symptoms (e.g. show intense interest in parts of objects) and ways of diagnosing ASD (e.g. there is currently no medical test to diagnose ASD). Knowledge about etiology (Harrison, Bradshaw et al., 2017) refers to the understanding of the causes of ASD

Six units that focus on early identification and intervention of ASD	Unit 1 Who is Khamdy? Unit 2 What are the early signs of ASD? Unit 3 What are the hindrances to diagnose ASD? Unit 4 Khamdy meets Dr. Xaiphone. Unit 5 Khamdy starts early intervention. Unit 6 Parent support group.
Six units that focus on social communication and inclusive strategies in a preschool setting.	Unit 7 Khamdy's new routine. Unit 8 Khamdy learns good behavior. Unit 9 Khamdy learns about preverbal communication. Unit 10 Khamdy's friends help him to communicate. Unit 11 Khamdy learns to talk. Unit 12 Khamdy learns to expand his language skills.
Eight units that focuses on differentiated instruction, inclusive and behavior management strategies in a primary school setting.	Unit 13 Khamdy goes to primary school. Unit 14 Khamdy learns about nonverbal communication. Unit 15 Khamdy learns to regulate his emotions. Unit 16 How can Khamdy develop executive functioning skills? Unit 17 Khamdy learns about friendship. Unit 18 Saying 'No' to bullying. Unit 19 Khamdy is becoming a teenager Unit 20 Khamdy is planning for his future.

Figure 1. Topics in the Story of Khamdy™ e-module (English version).

(e.g. autism is a brain-based disorder), and knowledge about treatment refers to treatment options (e.g. behavioral intervention is effective for children with ASD) and prognosis of ASD (e.g. children with ASD can grow up independently). Considering that cultural influences on ASD interpretation are widely established (e.g. Bernier et al., 2010; Burkett et al., 2015; Kang-Yi et al., 2018; Ratto et al., 2016), there is a basis to believe that the three dimensions of ASD knowledge as listed above are subjective to contextual and cultural influences, even though the neurobiological characteristics of ASD are universal and not culturally dependent.

The extent to which the behavioral expression of the neurobiological characteristics of ASD is interpreted can vary across cultures. For example, in Western cultures, speech delay and the lack of socialization are typically associated with initial concerns of ASD by parents and caretakers (Spikol et al., 2019). However, speech delay is less seen as an indication of a development concern in cultures that value behavioral compliance in young children, such as in Indian, Korean, or Chinese communities (Daley, 2004; Donohue et al., 2019; Kim, 2012; see La Roche et al., 2018 for a review). Without adequate knowledge to flag up early concerns about ASD, there will be an increased risk that young children with ASD in a particular community are not timely identified and diagnosed for the benefit of early intervention. This is a major concern in the developing and least developed countries, in which the lack of awareness and understanding about ASD symptoms among parents and early childhood educators has led to late identification of ASD beyond the critical period of early development (Bakare & Munir, 2011). In addition, knowledge about red flags of ASD and effective support strategies are critical for parents and teachers to make informed choices about intervention for their young children with ASD (Low et al., 2019).

The lack of these three aspects of knowledge is highly associated with stigma and misunderstanding about ASD intervention (Harrison, Slane et al., 2017; Kinnear et al., 2016). In low-resource contexts, access to diagnosis may be scarce (Barbaro & Halder, 2016), and parents and teachers have limited access to information about ASD,

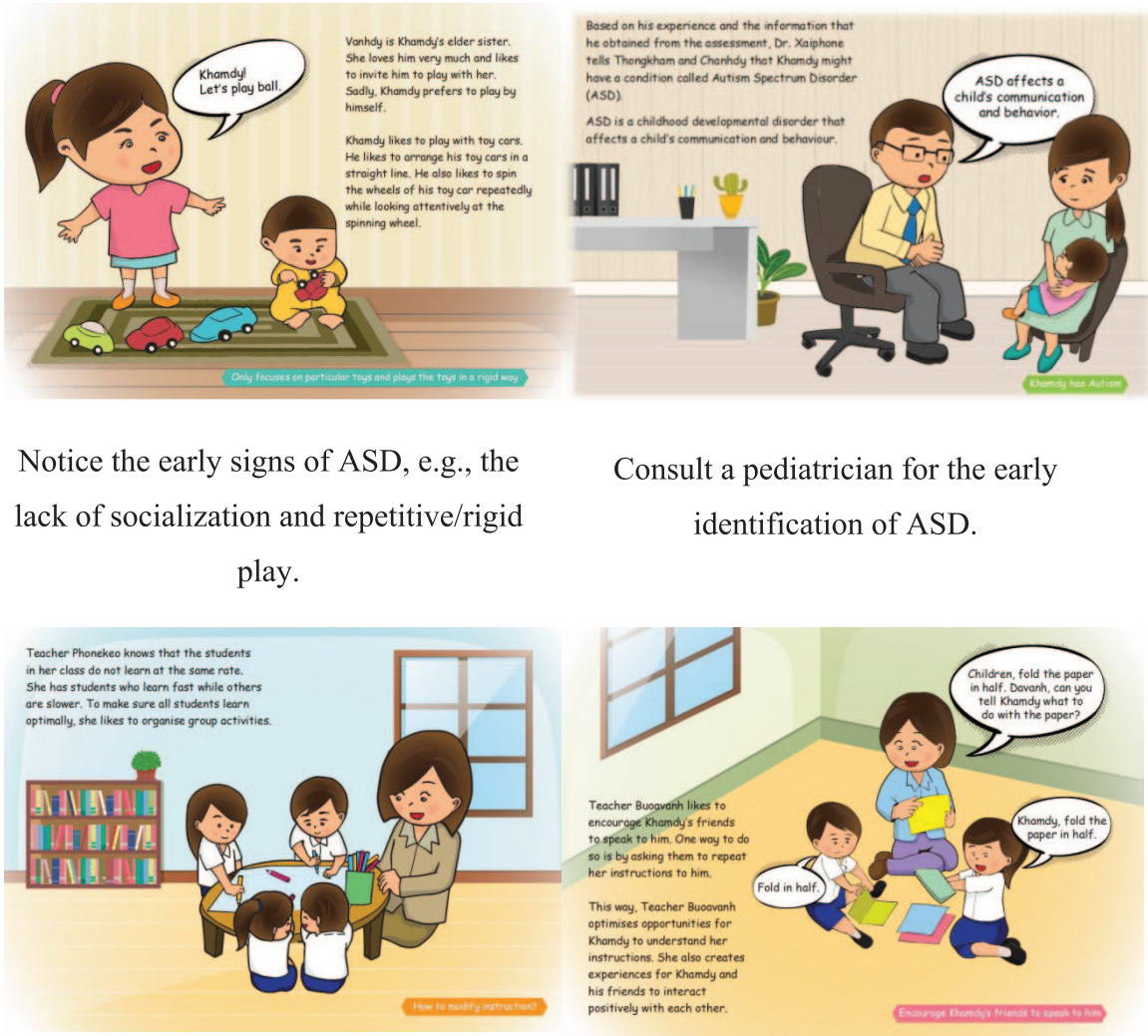

Figure 2. Selected screenshots in The Story of Khamdy™ e-module (English version).

including diagnosis and intervention (McCabe, 2013). As a result, they might form incomplete or imprecise understandings about ASD, leading to misconceptions and unrealistic expectations about the impact of ASD. In addition, the lack of knowledge about ASD intervention also makes it difficult for parents and teachers to evaluate program quality and be more thorough and persistent in a program intervention (McCabe, 2013). One way to overcome these issues is by making evidence-based, context-specific, and learner-centered ASD training available and accessible to parents and teachers. Preferably, such training would need to be locally designed and robustly evaluated to ensure its efficacy in facilitating adequate knowledge transfer among the major stakeholders of ASD care, support, and

intervention (Duggal et al., 2018). A locally contextualized and evidence-based ASD training program is particularly important for parents and teachers in the developing and least developed countries to combat stigma and misunderstanding about ASD (Duggal et al., 2018; Talahun et al., 2019). The Story of Khamdy e-module was developed by considering all factors as outlined above.

Social validity

Conventionally, social validity in education is used as a subjective measure of intervention acceptability. In other words, it is commonly used to capture the extent to which teachers and professionals perceive that an intervention is acceptable and appropriate to support children's needs that are otherwise marginalized and problematic (Berger et al., 2016; Elliott, 2017; King & Valdovinos, 2009). This includes the perceptual judgment of the fidelity of intervention goals, procedures, and targeted outcomes (Foster & Mash, 1999; King & Valdovinos, 2009). Other terms that are closely associated with "social validity" are "acceptability," "appropriateness," "fairness," and "reasonableness" of an intervention (Elliott, 2017).

Kim et al. (2017) expanded the use of social validity to evaluate how much paraprofessionals, who received training on behavioral support, perceived the relevance of the training in relation to enhancing positive behavioral changes in their students. By expanding the concept of "social validity" from "acceptability of intervention" to "acceptability of training," it provides insights into whether professional training has met its anticipated outcomes. In line with Kim et al. (2017), this study adopted "acceptability and appropriateness of training" as the key notion of social validity to evaluate the introduction of the pictorial narrative intervention program to a sample of primary school teachers in the Lao PDR.

Current study aims

The incorporation of pictorial narration in teacher training on ASD is considerably novel to teachers in the Lao PDR. The acceptability and appropriateness of this training approach, as perceived by the actual users, are pertinent to shed light on understanding the extent to which this training method can offer solutions in teacher training on effective inclusion for students with ASD. This training could potentially offer a solution for teacher training on ASD in low-resource settings in which the effective implementation of many conventional trainings is limited by language and cultural constraints. However, it remains unknown to what extent the local users, namely the targeted teachers, respond to this novel training approach. Specifically, it is intriguing to examine how the teachers feel about the training method (i.e. pictorial narrative approach) and content (i.e. the contextually developed storyline that incorporates evidence-based information). Therefore, it is important that these perceptual responses to the training were explored to disclose whether this novel ASD teacher training developed was meeting its original aim as aspired—to develop a culturally appropriate and contextually relevant ASD teacher training module that is effective and can be accepted by the local teachers in a low-resource context. Three research questions sought to be addressed in this study are as follows:

1. **Research Question 1:** Are there significant differences in knowledge change among the teachers after they participated in the ASD teacher training?
2. **Research Question 2:** Are there significant differences in stigma reduction among the teachers after they participated in the ASD teacher training?
3. **Research Question 3:** What is the social validity of using pictorial narration as a teacher training method for ASD?
4. **Research Question 4:** What are the associations between social validity, changes in ASD knowledge, and stigma as the effect of having had the training?

Method

The teacher training project was conducted in Luang Prabang, the Lao PDR. The project was designed according to the Action Design Research framework (Sein et al., 2011), in which a series of building-intervention-evaluation (BIE) cycles were conducted to produce the ASD teacher training module called The Story of Khamdy. The Action Design Research framework (Sein et al., 2011) was originally developed for the design and development of IT artifacts in industrial settings. This research framework was adopted in the project, as it prioritized the use of a prescriptive design knowledge through rigorous BIE cycles that reflect not only the theoretical precursors and intent of the researchers but also the actual needs of the users in the context (Sein et al., 2011). Therefore, efforts were made to ensure that local teacher trainers and teachers participated in each BIE cycle. The cross-country teamwork further enabled both cultural and functional equivalence (Jones et al., 2001) in the module developed.

The module was originally developed in English (https://www.facebook.com/groups/965835180415612/learning_content/) and then translated into the Lao language (https://www.facebook.com/groups/640036246483905/learning_content/). The translation team comprised two local teacher trainers, who were also involved in the development of the story plots. Both local teacher trainers have bachelor's degrees in the English language. In addition, one of them had attended training in inclusive education prior to joining this research project. The translation was carried out with the pictorial

illustrations as reference. The involvement of specialized experts and the availability of the pictorial illustrations had eased the translation process. The principles of cross-cultural translation and content validation proposed by Jones et al. (2001) and Squires et al. (2013) were followed during the translation process, with the exception that the back translation was not performed due to time and manpower constraints. The active participation of the local teacher trainers throughout the design, development, and translation process preserved the local values to this e-module so that the stories are both culturally and linguistically relevant to the local communities. The translated scripts with the pictorial illustrations were then used in a pilot study involving four Laotian student teachers in a local university in Luang Prabang, the Lao PDR. The student teachers were asked to identify those parts of the translated content that were unclear to them. After the pilot study, amendments were made to make the meanings more explicit for potential end-users.

Story building

The Story of Khamdy was based on the concept of pictorial narration. This concept emerged after an international team of researchers first visited Luang Prabang, Lao PDR, to conduct a needs assessment, together with local partners, at the beginning of the teacher training project. Luang Prabang is in north central Lao PDR. It consists of 58 villages, of which 33 comprise a United Nations Educational, Scientific and Cultural Organization (UNESCO) World Heritage Site. During the needs assessment, the international researchers soon realized that there are no local ASD-related knowledge resources and no existing vocabulary repertoires available to be used in the intended ASD teacher training; this confirmed the need to build a locally contextualized teacher training module on the effective inclusion of students with ASD.

To ensure that the module is relevant and pragmatic enough to the local teachers in Lao PDR, the team of module developers consisted of international researchers, who contribute knowledge about ASD, and local teacher trainers, who provided cultural insights, language translation, and editing advice throughout the course of the module development. The module built consisted of three main components: storyline, quiz, and assessment, which are compiled for public access in the Facebook social learning platform (https://www.facebook.com/groups/640036246483905/learning_content/).

Component 1: Story	A 20-unit story was developed and named The Story of Khamdy™. The story was first created in English and translated into the Lao language by the local teacher trainers in Luang Prabang. The story is represented by 20 videos in MP4 format. Units 1–6 are about early diagnosis and intervention. The story begins with the birth of Khamdy, a boy born in a happy and close-knit family, followed by the parents noticing developmental differences in Khamdy, and their experiences with ASD-related stigma and hindrances to getting an early diagnosis, and finally, to successfully obtaining an early diagnosis and support for early intervention. Units 7–12 are about Khamdy's experiences in preschool. A model teacher, called Teacher Bouvanh, is introduced. These units focus on how Teacher Bouvanh uses various social communication strategies to encourage Khamdy to develop speech, language, and communication skills in an inclusive preschool setting. Khamdy transforms from a minimally verbal child to being able to communicate his needs using different communication modes, including speech and augmentative and alternative communication (AAC). Units 13–20 are about Khamdy's development in primary school. Challenges related to the demands of primary education, such as transitioning, executive functions, friendship, bullying, and planning for the future, are highlighted. In these units, another model teacher, Teacher Phonekeo, is introduced. Teacher Phonekeo supports Khamdy with a wide range of differential instructional, inclusive, and behavior management techniques. Teacher Phonekeo also utilizes peer-mediated strategies and community education strategies to support Khamdy's learning in an inclusive primary school environment.
Component 2: Quiz	After watching the video of each unit of the story, the module learners will be instructed to answer five quiz questions about the video content. The quiz questions were designed to prompt information recall. Feedback is auto-generated in the online system to enable reflective learning.
Component 3: Assessment	After every five units, the module learners will have to complete an assessment that contains four new case scenarios. In each scenario, the module learners have to identify solutions based on knowledge learnt from the videos. The assessment requires skills related to knowledge application, analysis, and problem-solving.

Intervention

Following the development of the module, two teacher training sessions were conducted. The teacher trainings were conducted in a local information and communication technologies (ICT) center in Luang Prabang. A total of 112 teachers from 25 preschools and primary schools in the district of Luang Prabang participated in the teacher training

Unit 1:	2. Khamdy was diagnosed with Autism at birth. ○ True ○ False
Unit 2:	1. Khamdy _____ when his mother called his name. ○ cries ○ did not respond ○ smiles ○ looks up
Unit 3:	1. Will Khamdy became better if his parents just wait and see? ○ Yes ○ No

Figure 3. Selected quiz questions in The Story of Khamdy™ e-module (English version).

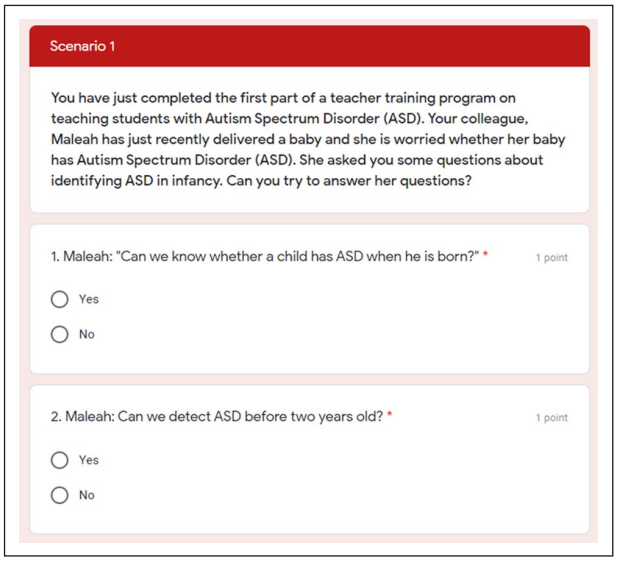

Figure 4. Selected assessment questions in The Story of Khamdy™ e-module (English version).

sessions. The teachers were divided into two groups (55 teachers each) because the computer room used for the orientation could only accommodate a maximum of 55 persons. Two trainings were carried out on 2 consecutive days, utilizing the same e-module and the same instructions. During the orientation session, the teachers were briefed about the e-module and were guided to perform the learning using the e-module. First, the videos from the first five units were shown using a projector in the training room. After the presentation of each video, the teachers were guided to answer quiz questions in the respective units (quizzes from units 1–5, refer to Figure 3), and after the presentation of the fifth video, they answered the first set of assessment questions (assessment I, refer to Figure 4), both using the paper-and-pen format. The paper-and-pen format was first used to familiarize the teachers with the storyline and module structure before they attempted the online version. This was used to reduce the teachers' cognitive load by having to attend to both new content and online navigation functions simultaneously. The instructor first familiarized them with the characters in the story and the associated tasks prior to switching to the online learning. After that, the teachers were introduced to the online version of the training materials, including access, videos, and the quizzes and assessments on a specially developed Khamdy Facebook social learning platform. The teachers were guided to answer the online quiz and assessment questions after the presentation of the next

three videos. Eight facilitators were present to assist the teachers and monitor their performance in the computer room. Upon ensuring that the teachers were able to use the online training platform independently, they were instructed to complete the remaining units at home within 4 weeks. They were encouraged to contact the Laotian facilitators, who acted as the admin in the Facebook social learning platform if they faced any technical issues. Our records indicated that no major issues were reported during this duration.

Evaluation

The module evaluation data were collected sequentially over the course of the teacher training via a one-sample, repeated-measure quasi-experimental design. Four instruments were used for the module evaluation, namely, personal information questionnaire, pre-training knowledge questionnaire, post-training knowledge questionnaire, and social validity questionnaire. These instruments are discussed in turn below.

Personal information questionnaire. When the teachers arrived on the day of training orientation, they were asked to fill out a questionnaire about their personal information (16 items). The personal information section consisted of questions related to age, gender, teacher experience, and their prior experience with students with ASD.

Pre- and post-training knowledge questionnaire (Autism Stigma and Knowledge Questionnaire). Before the training started, the teachers were instructed to fill in a questionnaire about their pre-training ASD knowledge, namely, the Lao version of the Autism Stigma and Knowledge Questionnaire (Lao ASK-Q). The Lao ASK-Q was translated from the original ASK-Q that Harrison, Bradshaw et al. (2017) developed. The Lao ASK-Q comprised 49 items (Low et al., 2021). The permission of translating ASK-Q into the Lao language for the purpose of this teacher training project was obtained from Harrison, Bradshaw et al. (2017).. Expert review and back translation measures were used to assess the validity of the translation, adhering to the cross-cultural translation recommendations outlined by Jones et al. (2001) and Squires et al. (2013). The translation process was explained in Low et al. (2021). After the teachers completed the entire training, that is, all 20 units, they were requested to fill in the Lao ASK-Q again but online. The pre- and post-training Lao ASK-Q comprised the same set of questions.

ASK-Q (Harrison, Bradshaw et al., 2017) consisted of 49 items that assessed three specific domains of ASD knowledge, namely, knowledge of diagnosis (18 items), etiology (16 items), and treatment (14 items), and a general question about ASD knowledge (i.e. "I have prior knowledge of autism"). In addition, Harrison, Bradshaw et al. (2017) had identified 7 items from the 49 item sets that reflect stigma (e.g. autism happens mostly in middle-class families). In the current study, to minimize the chances of having the teachers guess the answers, three categorical choices were provided in the "ASK knowledge" section: "Yes," "No," and "Not sure." In the written instruction, the teachers were reminded that whenever they were unsure of the answers, they could select "Not sure."

Among the 48 items about the specific domain knowledge in ASK-Q, there were 29 correct statements (e.g. some children with autism may lose acquired speech) and 19 incorrect statements (e.g. autism is preventable). Reserve coding was conducted for items with incorrect statements using the Excel template created by Harrison, Bradshaw et al. (2017). From this data cleaning procedure, only a unitary set of binary responses (i.e. "1" and "0") was produced, in which "1" reflected having had the correct knowledge, while "0" reflected not having had the correct knowledge or "Not sure." In short, each teacher would obtain one mark for a correct answer in ASK-Q. Composite scores were computed to represent their overall ASD knowledge and dimensional scores. After the teachers completed the entire training, that is, all 20 units, they were requested to fill in the post-training ASD knowledge questionnaire online. The post-training ASD knowledge questionnaire was the same set of questions translated from ASK-Q. Internal consistency was computed for the Lao ASK-Q used in this study. Alpha values of 0.94 and 0.87 were obtained for the pre- and post-training ASK-Q evaluations, respectively, reflecting good content validity. The internal consistency computed for the sub-scales in ASK-Q also indicated good content validity: Knowledge of Diagnosis ($\alpha_{pre}=0.89$, $\alpha_{post}=0.75$), Knowledge of Etiology ($\alpha_{pre}=0.86$, $\alpha_{post}=0.58$), Knowledge of Treatment ($\alpha_{pre}=0.80$, $\alpha_{post}=0.56$), and Stigma ($\alpha_{pre}=0.73$, $\alpha_{post}=0.61$).

Social validity questionnaire. In addition, the teachers were also requested to fill in a feedback form, which consisted of 27 questions concerning the social validity of the training. The social validity measured in this study encompassed two aspects, that is, social validity on training method (Part I) and social validity on training content (Part II). The social validity on training method (Part I) was measured using the adapted Primary Intervention Rating Scale (PIRS) developed by Lane et al. (2009). There were a total of 17 items in Part I of the social validity questionnaire, which aimed to measure the teachers' perception toward the design of the training procedure (i.e. the pictorial narrative enhanced e-module training method). PIRS was originally developed by Lane et al. (2009) to measure the social validity of school-wide positive behavior support intervention. Therefore, the word "intervention" was changed to "training" for the 17 items in Part 1 of the social validity questionnaire.

Part II consisted of 10 items from the social validity questionnaire developed by King and Valdovinos (2009).

This part aimed to measure the teachers' perception toward the content of the pictorial narration (i.e. the pedagogical method used by the teachers). The social validity questionnaire was originally developed by King and Valdovinos (2009) to measure the social validity of the content in an episode of "Super Nanny," a popular American television show in which pivotal response treatment was used to manage the behavior of a child with ASD. The original version of King and Valdovinos' (2009) social validity questionnaire consisted of 20 items, which asked questions related to the adults' behavior, the child's behavior, and the overall impression. To match with the purpose of the current investigation, only 10 items directly related to adults' behaviors were adopted and adapted. Adaptations made include changing the word "parents" to "teachers." The social validity questionnaire used in this study was also translated using the same process for the translation of ASK-Q.

Results

Although 112 teachers participated in the ASD teacher training, only 87 of them submitted all the required documents upon the completion of the ASD e-module 4 weeks later, which include the demographic information questionnaire, pre-training knowledge questionnaire, post-training questionnaire, and the social validity feedback form. This resulted in a response rate of 77.7%.

Teachers' profiles

Table 1 displays the demographic information of the teachers ($N=87$). They are predominantly represented by female teachers (80.5%). The majority of teachers can be considered as senior teachers because 74.7% of them were 30 years old and above, and 86.2% of them had more than 5 years of teaching experience, predominantly with students who are below 10 years of age (86.2%). Most of the teachers only had academic qualifications up to certificate (82.8%) and high-school (3.4%) levels.

In the demographic information questionnaire, the teachers were asked about their personal experience with ASD. Although more than 57.5% of them had heard about ASD, only 9.9% of them claimed that they had prior knowledge about ASD, and only 23.3% of them reported that they knew how to teach students with ASD. It can be deduced from the findings that at least more than half of the Laotian teachers in this study were aware of ASD. However, their knowledge and skills to teach students with ASD were reportedly low (refer to Figure 5).

Baseline correlations were conducted to assess the potential associations between demographic variables, knowledge of ASD (pre- and post-training), social validity of training method, and social validity of training content. The results from Pearson correlation analyses revealed insignificant correlations between all variables, except weak correlations between pre-training knowledge scores and teachers' qualifications ($r=0.204$, $p<0.05$), teaching experience ($r=-0.244$, $p<0.01$), the age group of students taught ($r=0.290$, $p<0.01$), knowing the term "autism" ($r=0.281$, $p<0.05$), having prior knowledge of ASD, ($r=0.406$, $p<0.01$), and knowing how to teach students with ASD ($r=0.367$, $p<0.01$). Considering that all the post-training scores (i.e. the post-training ASD knowledge scores and social validity scores) were not correlated with all other variables, we argued that the post-training scores genuinely reflect the effects of the training provided.

Table 1. Demographic information.

	n	Percent
Gender		
Male	17	19.5
Female	70	80.5
Age (years)		
20–29	22	25.3
30–39	28	32.2
40–49	31	35.6
50–59	6	6.9
Qualification		
High school	3	3.4
Certificate	72	82.8
Undergraduate degree	12	13.8
Teaching experience (years)		
1–5	12	13.8
6–10	15	17.2
11–15	15	17.2
16–20	22	25.3
21 and above	23	26.4
Age group of students (years)		
3–6	18	20.7
7–10	57	65.5
11–14	7	8.0
15–18	3	3.4
19 and above	2	2.3

ASD stigma and ASD knowledge

Paired-sample *t* test from pre- and post-training Lao ASK-Q results ($N=87$) revealed statistically significant knowledge gain in all three knowledge dimensions measured with ASK-Q (Harrison, Bradshaw et al., 2017), $p<0.001$ (refer to Figure 5). To ease data presentation, Harrison, Bradshaw et al. (2017) suggested a classification of knowledge levels measured by ASK-Q. The knowledge levels were differentiated into inadequate and adequate knowledge levels based on statistically pre-determined sub-score ranges (refer to Figure 6). Inadequate knowledge was associated with sub-score ranges of 0–10 for diagnosis and etiology knowledge dimensions, and 0–9 for treatment knowledge dimension. Adequate knowledge was associated with sub-score ranges 11–18 for diagnosis, 11–16 for etiology, and 10–14 for

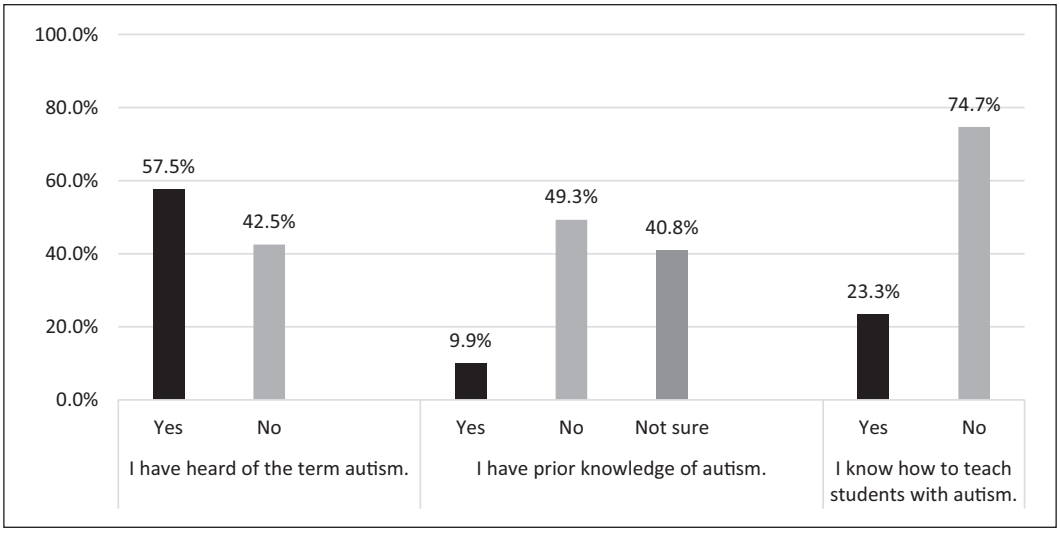

Figure 5. Personal experience with ASD.

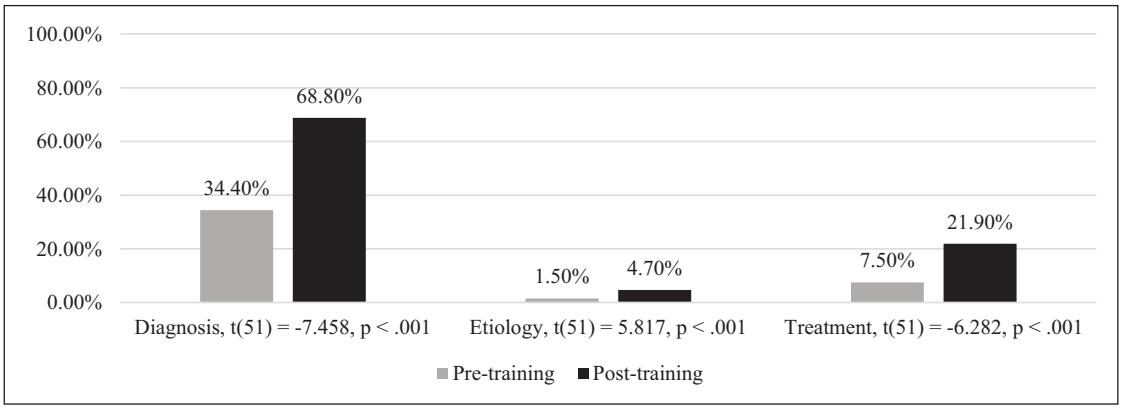

Figure 6. Pre- and post-training adequate knowledge levels for the three knowledge sub-scales.

treatment knowledge dimensions. Specifically, the most notable knowledge gain was recorded for the dimension of diagnosis: from only 34.3% of teachers having had adequate diagnosis-related knowledge before the ASD teacher training to as many as 68.8% of them having had adequate diagnosis-related knowledge after the training (34.5% increase). This was followed by increases in the dimension of treatment-related knowledge (from 7.5% with adequate knowledge to 21.9% with adequate knowledge), and the dimension of etiology-related knowledge (from 1.4% with adequate knowledge to 7.5% with adequate knowledge).

Paired-sample t test from pre- and post-training ASK-Q results (N=87) revealed statistically significant changes in ASD-related stigma measured with the Lao ASK-Q (Low et al., 2021), $p<0.001$ (refer to Figure 7). Similar to the measurement of ASD knowledge, Harrison, Bradshaw et al. (2017) also provided a classification of stigma endorsement measured by ASK-Q. The sub-score range of 0–2 was regarded as endorsing stigma, while the sub-score range of 3–7 was regarded as not endorsing stigma. The result revealed that there was a significant reduction in stigma, $t(51)=-7.590$, $p<0.001$. The ASK-Q results disclosed that as many as 78.1% of teachers did not endorse ASD-related stigma after completing the training, compared to only 45.1% of them who did not endorse stigma before the training.

Social validity of the training method

For social validity of the training method, as measured using the adapted PIRS, the cumulative mean was 5.45 (standard deviation=0.420), indicating a high social validity rating on a 6-point Likert-type scale. The means for all the items are presented in Figure 8 below. Specifically, most teachers felt that the training was a good way to support teachers to teach students with ASD in schools (M=5.79, σ=0.411), especially teachers in primary schools (M=5.74, σ=0.518). Therefore, they were keen to suggest the use of this training to other teachers (M=5.74, σ=0.444), as they felt that the

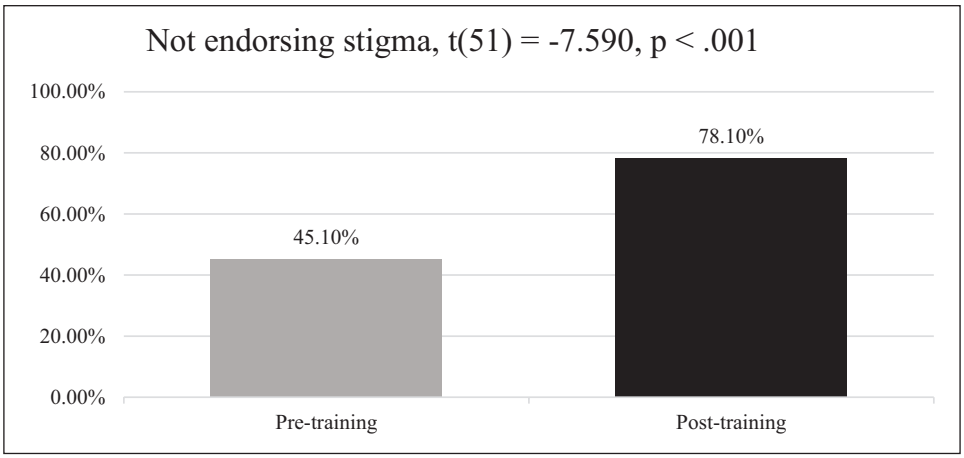

Figure 7. Pre- and post-training levels of not endorsing stigma.

training was a fair way to train teachers on how to teach students with ASD ($M = 5.72$, $\sigma = 0.491$) and they liked the procedures used in this training ($M = 5.72$, $\sigma = 0.491$). However, it is worth noticing that a lower mean was recorded for the fact that the strategies in this training are consistent with those they have used in schools ($M = 5.37$, $\sigma = 0.858$).

Social validity of the training content

For social validity of the training content, as measured using the social validity questionnaire developed by King and Valdovinos (2009), the cumulative mean was 5.62 (standard deviation = 0.341), indicating a high social validity rating on a 6-point Likert-type scale. The means for all the items are presented in Figure 9 below. Specifically, five items were recorded with means above the cumulative mean, items 10, 1, 2, 5, and 7 (range: 5.68–5.79). Specifically, the teachers were particularly in favor of the fact that the teachers in The Story of Khamdy treat their students humanely ($M = 5.79$, $\sigma = 0.411$). Besides that, they liked the methods in which the teachers in the story dealt with Khamdy's behaviors ($M = 5.74$, $\sigma = 0.552$), which they believed would produce a lasting improvement in Khamdy's social interactions ($M = 5.72$, $\sigma = 0.491$). Consequently, the teachers felt that the ways the teachers in the story interacted with Khamdy were worth the time and effort ($M = 5.72$, $\sigma = 0.453$), as those effects are sensitive to students' learning needs ($M = 5.68$, $\sigma = 0.469$). Despite a generally positive response, it is worth noticing that the teachers were relatively unsure whether most teachers would be able to apply similar interaction methods with their students with ASD ($M = 5.39$, $\sigma = 0.526$).

Associations between social validity and the changes in ASD knowledge and stigma

One-tailed correlations were conducted to examine the associations between social validity and the changes in ASD knowledge and stigma (i.e. pre- and post-training score differences) as the effect of having had the training. The results revealed significant correlation between training acceptance and changes in the total knowledge scores, $r = 0.269$, $p < 0.05$ (Table 2). More specifically, training acceptance was found to relate to changes in diagnosis-related ($r = 0.256$, $p < 0.05$) and etiology-related knowledge ($r = 0.267$, $p < 0.05$) (Table 2).

Discussion

The main aim of this article was to explore whether pictorial narration can offer a solution to teacher training on the effective inclusion of students with ASD in a low-resource context in the Lao PDR. For this purpose, pre- and post-training knowledge data were successfully collected from 87 teachers who participated in the teacher training. In this article, two sets of data were presented: (1) the teachers' ASD knowledge and stigma change and (2) the social validity of the training. First of all, the comparison of pre- and post-training knowledge results revealed statistically significant changes in all knowledge dimensions, as measured by the Lao ASK-Q (Low et al., 2021). The highest level of improvement was seen in the teachers' diagnosis-related knowledge.

The findings suggest that the Laotian teachers had improved in the ability to recognize and differentiate ASD symptoms upon participating in the training that was designed using the pictorial narrative approach. Beyond the context of health communication, the current findings extend the effectiveness of using pictorial communication for the purpose of teacher training on ASD in a low-resource context. In particular, the use of pictorial and audiovisual communication is believed to enhance message retention (Muturi, 2016) and recall (van Weert et al., 2011) among the teachers. In the current study, the number of teachers who gained adequate ASD diagnosis-related knowledge after the training (68.8%) doubled the number

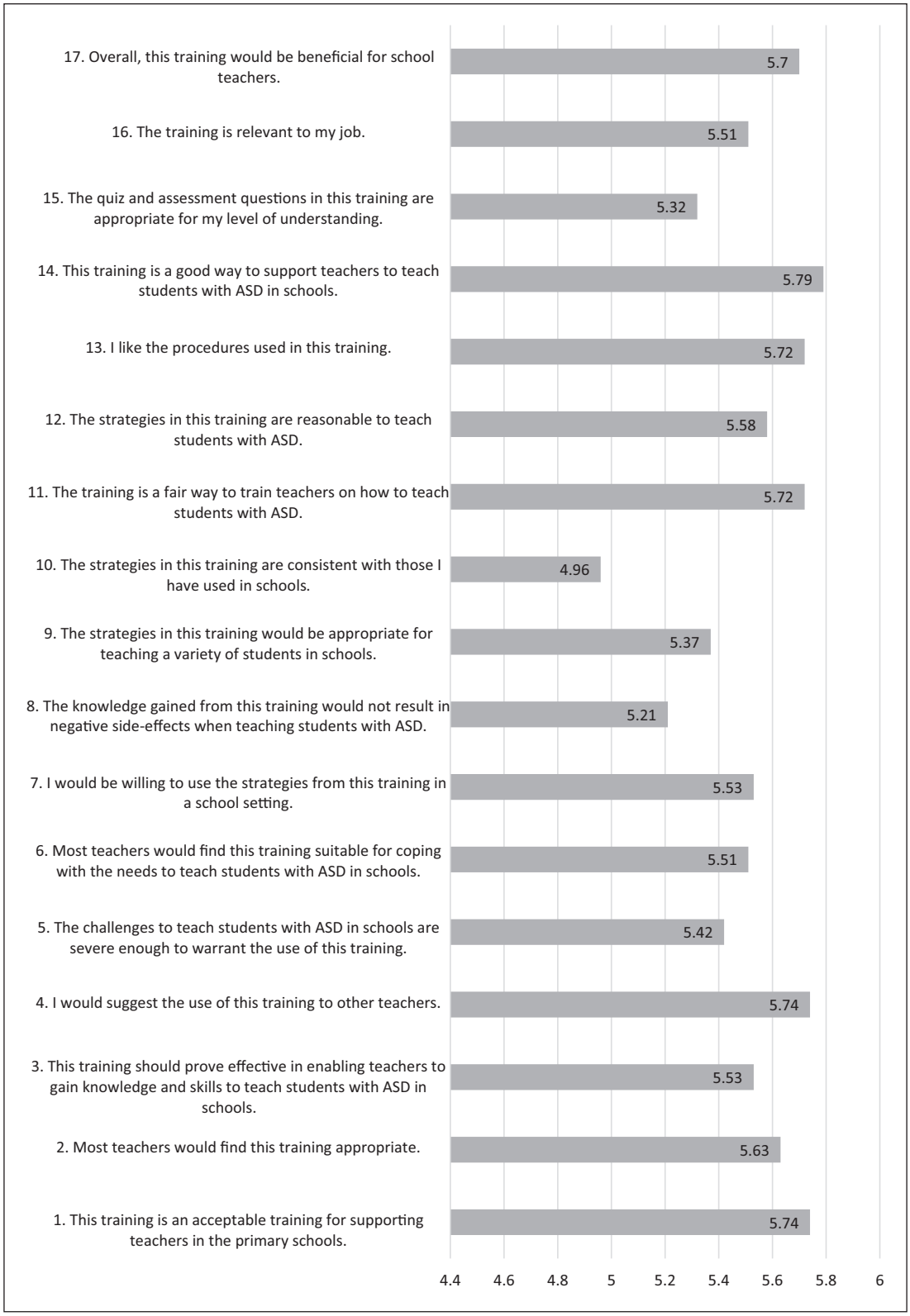

Figure 8. Means of social validity of training method.

of teachers with adequate ASD diagnosis-related knowledge before the training (34.3%). In fact, the recorded ASD diagnosis-related knowledge of the Laotian teachers after the training exceeded the knowledge levels recorded from other international communities such as China, 65.0%, from Yu et al. (2020), and the global norm of

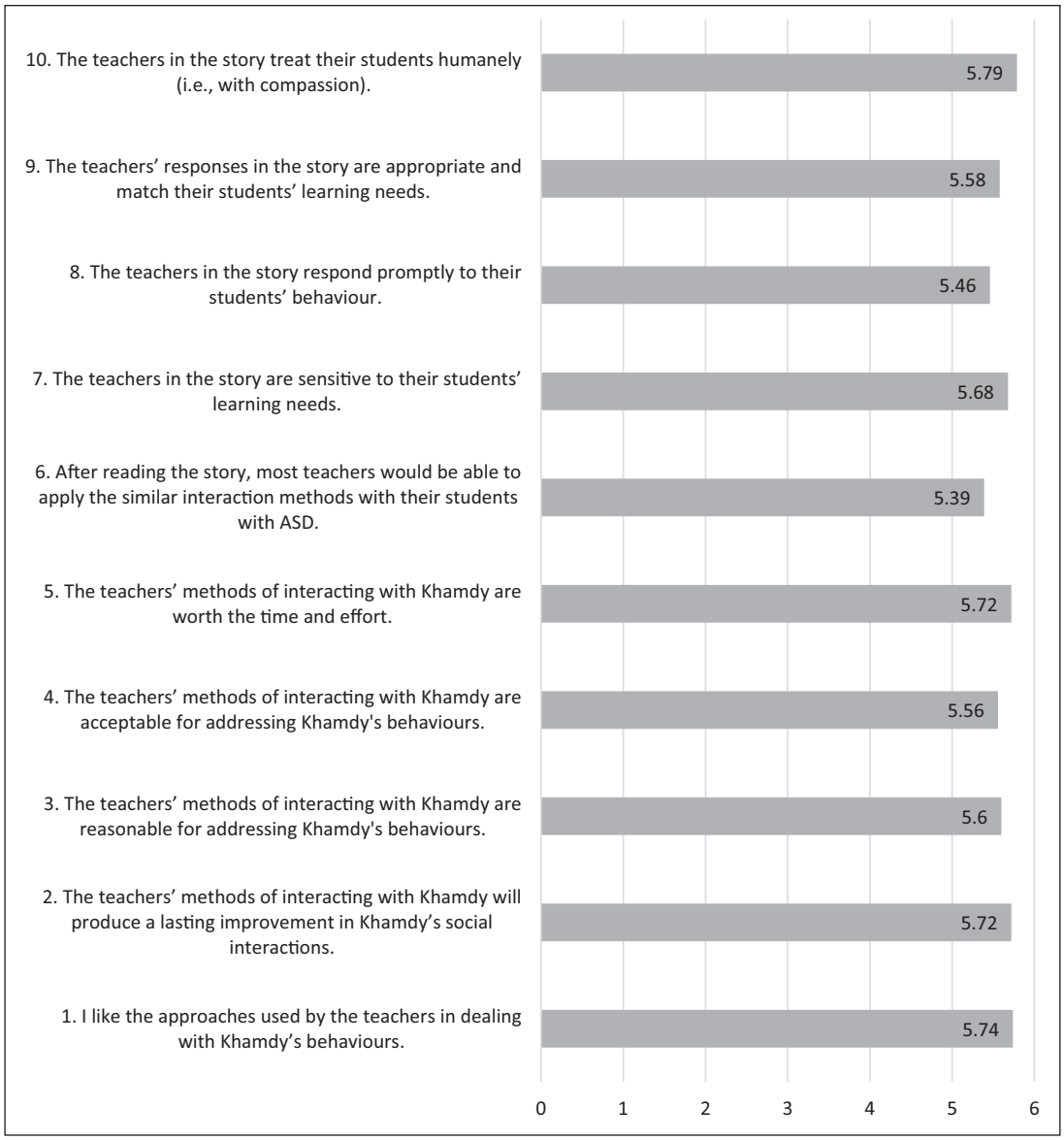

Figure 9. Means of social validity of training content.

approximately 72% established by Harrison, Slane et al. (2017). This set of findings indicates that it is possible to close the ASD knowledge gap of teachers in a low-resource context through a well-validated teacher training program as implemented in this study.

Significant changes in etiology-related (from 1.4% with adequate knowledge to 7.5% with adequate knowledge) and treatment-related knowledge on ASD (from 7.5% with adequate knowledge to 21.9% with adequate knowledge) were also recorded in the Laotian teachers. It is noteworthy that these were fivefold and threefold increases in both dimensions, albeit still lower than the global standard of 55.8% for etiology-related knowledge and 56.9% for treatment-related knowledge that Harrison, Slane et al. (2017) recorded. Again, the findings supported the efficacy of the use of a systematic and elaborated pictorial narrative teacher training program in driving positive gains in all three aspects of ASD knowledge measured in this study.

At the same time, the findings pointed to a higher degree of challenge in imparting etiology- and treatment-related knowledge than imparting diagnosis-related knowledge via the ASD teacher training e-module that was developed. There are two possible reasons. First is the training dosage effect (Reyes et al., 2012). Higher doses of training (i.e. more training sessions or longer duration of training) are possibly required to produce more optimal results in training related to the understanding of ASD etiology and treatment. The teacher training implemented in this study only spanned across a 4-week duration. It is unrealistic to expect a drastic shift in etiology- and

Table 2. Correlation results.

	Mean scores of training acceptance	Mean scores of content acceptance
Score differences (Diagnosis, D)		
Pearson correlation	0.256*	0.185
Sig. (one-tailed)	0.040	0.104
Score differences (Etiology, E)		
Pearson correlation	0.267*	0.103
Sig. (one-tailed)	0.033	0.242
Score differences (Treatment, T)		
Pearson correlation	0.100	0.083
Sig. (one-tailed)	0.250	0.289
Score differences (Stigma)		
Pearson correlation	0.142	0.221
Sig. (one-tailed)	0.160	0.060
Score differences (Knowledge, D + E + T)		
Pearson correlation	0.269*	0.162
Sig. (one-tailed)	0.031	0.133

*Correlation is significant at the 0.05 level (one-tailed).

treatment-related knowledge within this short amount of time, considering that a prominent majority of the Laotian teachers (over 90%) commenced the training with inadequate etiology- and treatment-related knowledge prior to the ASD teacher training.

The above findings pointed to the differences between fact- and skill-based knowledge. It is possible that fact-based knowledge (such as knowledge about ASD symptoms and diagnostic procedures) can be easily acquired by the knowledge users in a small amount of time, as compared to skill-based knowledge (such as knowledge about the planning and execution of ASD-related treatment). Second, it is also possible that the current model of teacher training is effective in imparting fact-based knowledge, but perhaps less so for skill-based knowledge, which generally requires the reinforcement of learning via practical experience. The findings pointed to the necessity to augment the current model of teacher training by adding a hands-on component to reinforce the acquisition of treatment-related knowledge. Possibly, practice-oriented workshops can be incorporated in the future to fill in this gap.

In conjunction with the increase in ASD knowledge, the findings indicated a parallel reduction in ASD-related stigma among the teachers who had the training. Considering that stigma is a major hindrance in getting proper diagnosis and intervention (Bakare et al., 2019), particularly in many culturally exclusive and remote communities in low-resource countries, the reduction of stigma recorded is directly indicative of the values of the intended training in reducing stigma among the teachers in Luang Prabang. There were fewer teachers who endorsed stigma (only 21.9%) compared to as many as 54.5% of them prior to the teacher training. As advocated by past researchers (such as Duggal et al., 2018; Talahun et al., 2019), a locally contextualized and evidence-based ASD training program is important for parents and teachers in the developing and least developed countries to combat stigma and misunderstanding about ASD. The current finding offers solid evidence to support this advocacy.

The second aim of this article was to explore the social validity of training. Considering the use of a pictorial narrative enhanced e-module learning was relatively novel in this low-resource setting, so two aspects of social validity were explored, namely, social validity of training method (i.e. pictorial narration in an e-learning format) and social validity of training content (i.e. the construct of the story), which encompasses the pedagogical approach recommended for teachers for the teaching of students with ASD in an inclusive environment. The findings pointed to an overwhelmingly positive rating in these two aspects of social validity. A cumulative mean of 5.45 (6-point scale) was recorded for social validity of training method, and a cumulative mean of 5.62 (6-point scale) was recorded for social validity of training content. Both findings reinforced high acceptance toward the use of a cultural-oriented pictorial narrative approach in ASD teacher training. The findings supported the feasibility of using this approach for the purpose of teacher training in the area of ASD.

Despite the risk of social desirability bias in the reporting, there are reasons to believe that, largely, the rating is genuine. First, high ratings were noted for items related to the relevance of the training to the teachers in Luang Prabang. This shows that the development of a teacher training resource in which names and characteristics of the characters were reflective of the Lao's context was rated highly by the local teachers in the current research setting. Second, in comparison with other responses, the teachers were relatively more doubtful about the effects of the teaching strategies

recommended in the training materials. Presumably, some of the inclusive teaching strategies recommended were novel to the current practices in schools in the Lao PDR. As a result, they were unsure whether the teaching strategies would produce negative side effects in the children (item 8 in Figure 8). Even so, the teachers reported high ratings in social validity on the training content. Their overall ratings pointed to high acceptance toward the methods used by the fictional teachers in the story.

Last, but not least, statistically significant correlation was found in the teachers' acceptance of training and their knowledge gain, particularly in the domains of diagnosis- and etiology-related knowledge. The findings validated the application of a pictorial narrative teacher training method in steering positive knowledge change among the teachers. The findings supported the notion that culturally remote communities are inclined to use arts, pictures, and oral communication as the methods of passing on knowledge in the community (Geana et al., 2012). The findings on social validity in this study supported this notion. As found in this study, the local Laotian teachers reported positive perceptions toward the pictorial narrative method of teacher training. More importantly, their positive perception toward the training method was found to correlate with their knowledge gain, pointing to the efficacy of a pictorial narrative method of teacher training as a solution to transfer ASD knowledge to a low-resource setting like the Lao PDR.

As for the limitation, it is important to recognize that the data were collected from 87 of 112 teachers who participated in the teacher training; 77.7% was a notably good response rate, considering that this was the first online teacher training attempted for the purpose of professional teacher development on the topic of inclusive education and ASD in the Lao PDR. However, at this same time, it also revealed the fact that not all teachers were proficient in the use of technology. Some of the teachers might have faced Internet connectivity issues at home. Given this finding, it is important to recognize the individual needs of all teachers, including those who were residing in remote areas. The next phase of the project will aim to address this by developing an off-line version of the pictorial narrative training materials. In addition, considering that training of trainers (ToT) has been recognized as a strong predictor of sustainability in knowledge transfer (Mormina & Pinder, 2018), the next phase of the project will also focus on the training of local trainers to strengthen the local peer-to-peer support on the topic of inclusive education and ASD in the Lao PDR. This step is recommended to ensure continuity in knowledge mobilization and the development of local resources and supports in a more sustainable way.

Conclusion

Overall, the findings from this study supported the application of pictorial narration for the purpose of ASD teacher training in low-resource settings. The current findings extend the application of pictorial communication in teacher training on the topic of ASD, delivered through a storyline with names and characteristics of the characters that were reflective of the local culture.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This project is supported by a research funding from The HEAD Foundation (https://headfoundation.org/) [304/PGURU/650952/T135].

ORCID iDs

Hui Min Low https://orcid.org/0000-0002-8595-9215

Tze Peng Wong https://orcid.org/0000-0001-6593-2309

Ann Sien Sut Lee https://orcid.org/0000-0001-7806-9911

References

Abma, T. A., Cook, T., Rämgård, M., Kleba, E., Harris, J., & Wallerstein, N. (2017). Social impact of participatory health research: Collaborative non-linear processes of knowledge mobilization. *Educational Action Research*, *25*(4), 489–505.

Bakare, M. O., & Munir, K. M. (2011). Excess of non-verbal cases of autism spectrum disorders presenting to orthodox clinical practice in Africa–a trend possibly resulting from late diagnosis and intervention. *South African Journal of Psychiatry*, *17*(4), 118–120.

Bakare, M. O., Taiwo, O. G., Bello-Mojeed, M. A., & Munir, K. M. (2019). Autism Spectrum Disorders in Nigeria: A scoping review of literature and opinion on future research and social Policy Directions. *Journal of Health Care for the Poor and Underserved*, *30*(3), 899–909.

Barbaro, J., & Halder, S. (2016). Early identification of autism spectrum disorder: Current challenges and future global directions. *Current Developmental Disorders Reports*, *3*(1), 67–74.

Berger, N. I., Manston, L., & Ingersoll, B. (2016). Establishing a scale for assessing the social validity of skill building interventions for young children with autism spectrum disorder. *Journal of Autism and Developmental Disorders*, *46*(10), 3258–3269.

Bernier, R., Mao, A., & Yen, J. (2010). Psychopathology, families, and culture: Autism. *Child and Adolescent Psychiatric Clinics*, *19*(4), 855–867.

Bresciani, S., Arora, P., & Kernbach, S. (2018, July). Education and culture affect visualization's effectiveness for health communication. In *2018 22nd international conference information visualisation (IV)* (pp. 386–389). IEEE.

Brewer, N. T., Hall, M. G., Noar, S. M., Parada, H., Stein-Seroussi, A., Bach, L. E., . . . Ribisl, K. M. (2016). Effect of pictorial cigarette pack warnings on changes in smoking behavior: A randomized clinical trial. *JAMA Internal Medicine*, *176*(7), 905–912.

Burkett, K., Morris, E., Manning-Courtney, P., Anthony, J., & Shambley-Ebron, D. (2015). African American families on autism diagnosis and treatment: The influence of culture. *Journal of Autism and Developmental Disorders*, *45*(10), 3244–3254.

Clawson, T. H., Leafman, J., Nehrenz Sr, G. M., & Kimmer, S. (2012). Using pictograms for communication. *Military Medicine*, *177*(3), 291–295.

Colledge, A., Car, J., Donnelly, A., & Majeed, A. (2008). Health information for patients: Time to look beyond patient information leaflets. *Journal of the Royal Society of Medicine*, *101*(9), 447–453.

Daley, T. C. (2004). From symptom recognition to diagnosis: Children with autism in urban India. *Social Science & Medicine*, *58*(7), 1323–1335.

Donohue, M. R., Childs, A. W., Richards, M., & Robins, D. L. (2019). Race influences parent report of concerns about symptoms of autism spectrum disorder. *Autism*, *23*(1), 100–111.

Duggal, C., Dalwai, S., & Datta, V. (2018). Diagnosis and interventions for autism spectrum disorder in a multidisciplinary child development centre in India. *Journal of Psychology & Clinical Psychiatry*, *9*(6), 660–664.

Edward, J., Morris, S., Mataoui, F., Granberry, P., Williams, M. V., & Torres, I. (2018). The impact of health and health insurance literacy on access to care for Hispanic/Latino communities. *Public Health Nursing*, *35*(3), 176–183.

Elleström, L. (2019). *Transmedial narration: Narratives and stories in different media*. Palgrave Pivot.

Elliott, S. N. (2017). The social validity of "Acceptability of behavioral interventions used in classrooms": Inferences from longitudinal evidence. *Behavioral Disorders*, *43*(1), 269–273.

Foster, S. L., & Mash, E. J. (1999). Assessing social validity in clinical treatment research: Issues and procedures. *Journal of Consulting and Clinical Psychology*, *67*(3), 308.

Geana, M. V., Greiner, K. A., Cully, A., Talawyma, M., & Daley, C. M. (2012). Improving health promotion to American Indians in the Midwest United States: Preferred sources of health information and its use for the medical encounter. *Journal of Community Health*, *37*(6), 1253–1263.

Harrison, A. J., Bradshaw, L. P., Naqvi, N. C., Paff, M. L., & Campbell, J. M. (2017). Development and psychometric evaluation of the autism stigma and knowledge questionnaire (ASK-Q). *Journal of Autism and Developmental Disorders*, *47*(10), 3281–3295.

Harrison, A. J., Slane, M. M., Hoang, L., & Campbell, J. M. (2017). An international review of autism knowledge assessment measures. *Autism*, *21*(3), 262–275.

Jones, P. S., Lee, J. W., Phillips, L. R., Zhang, X. E., & Jaceldo, K. B. (2001). An adaptation of Brislin's translation model for cross-cultural research. *Nursing Research*, *50*(5), 300–304.

Kang-Yi, C. D., Grinker, R. R., Beidas, R., Agha, A., Russell, R., Shah, S. B., . . . Mandell, D. S. (2018). Influence of community-level cultural beliefs about autism on families' and professionals' care for children. *Transcultural Psychiatry*, *55*(5), 623–647.

Kaur, N. (2019). Impact of photonovel interventions among people with low levels of health literacy: A literature review. *Education for Information*, *35*(1), 21–34.

Kim, H. U. (2012). Autism across cultures: Rethinking autism. *Disability & Society*, *27*(4), 535–545.

Kim, S., Koegel, R. L., & Koegel, L. K. (2017). Training paraprofessionals to target socialization in students with ASD: Fidelity of implementation and social validity. *Journal of Positive Behavior Interventions*, *19*(2), 102–114.

King, M. J., & Valdovinos, M. G. (2009). Social validity of behavioral practices in the treatment of autism—A review of the Super Nanny. *Research in Autism Spectrum Disorders*, *3*(1), 173–178.

Kinnear, S. H., Link, B. G., Ballan, M. S., & Fischbach, R. L. (2016). Understanding the experience of stigma for parents of children with autism spectrum disorder and the role stigma plays in families' lives. *Journal of Autism and Developmental Disorders*, *46*(3), 942–953.

Koops van't Jagt, R., de Winter, A. F., Reijneveld, S. A., Hoeks, J. C., & Jansen, C. J. (2016). Development of a communication intervention for older adults with limited health literacy: Photo stories to support doctor–patient communication. *Journal of Health Communication*, *21*(sup2), 69–82.

Lane, K. L., Kalberg, J. R., Bruhn, A. L., Driscoll, S. A., Wehby, J. H., & Elliott, S. N. (2009). Assessing social validity of school-wide positive behavior support plans: Evidence for the reliability and structure of the Primary Intervention Rating Scale. *School Psychology Review*, *38*(1), 135–144.

La Roche, M. J., Bush, H. H., & D'Angelo, E. (2018). The assessment and treatment of autism spectrum disorder: A cultural examination. *Practice Innovations*, *3*(2), 107–122.

Low, H. M., Lee, L. W., & Che Ahmad, A. (2019). Knowledge and attitudes of special education teachers towards the inclusion of students with autism spectrum disorder. *International Journal of Disability, Development and Education*, *67*, 497–514.

Low, H. M., Wong, T. P., Lee, L. W., Makesavanh, S., Vongsouangtham, B., Phannalath, V., San, S., Che Ahamd, A., & Lee, A. S. S. (2021). Brief report: A grassroot investigation of ASD knowledge and stigma among teachers in Luang Prabang, Lao PDR. *Research in Autism Spectrum Disorder*, *80*, 101694.

McCabe, H. (2013). Bamboo shoots after the rain: Development and challenges of autism intervention in China. *Autism*, *17*(5), 510–526.

Meindl, J. N., Delgado, D., & Casey, L. B. (2020). Increasing engagement in students with autism in inclusion classrooms. *Children and Youth Services Review*, *111*, 104854. https://doi.org/10.1016/j.childyouth.2020.104854

Meppelink, C. S., Smit, E. G., Diviani, N., & Van Weert, J. C. (2016). Health literacy and online health information processing: Unraveling the underlying mechanisms. *Journal of Health Communication*, *21*(Suppl. 2), 109–120.

Mormina, M., & Pinder, S. (2018). A conceptual framework for training of trainers (ToT) interventions in global health. *Globalization and Health*, *14*(1), 1–11.

Muturi, N. (2016). Community perspectives on communication strategies for alcohol abuse prevention in rural central Kenya. *Journal of Health Communication*, *21*(3), 309–317.

Paivio, A. (1991). Dual coding theory: Retrospect and current status. *Canadian Journal of Psychology/Revue Canadienne de Psychologie*, *45*(3), 255.

Parsons, S., & Kovshoff, H. (2019). Building the evidence base through school-research partnerships in autism education: The Autism Community Research Network@ Southampton [ACoRNS]. *Good Autism Practice*, *20*(1), 5–12.

Prince, G. (2012). *Narratology: The form and functioning of narrative*. Mounton Publishers.

Quaglio, G., Sørensen, K., Rübig, P., Bertinato, L., Brand, H., Karapiperis, T., . . . Dario, C. (2017). Accelerating the health literacy agenda in Europe. *Health Promotion International*, *32*(6), 1074–1080.

Ratto, A. B., Reznick, J. S., & Turner-Brown, L. (2016). Cultural effects on the diagnosis of autism spectrum disorder among Latinos. *Focus on Autism and Other Developmental Disabilities*, *31*(4), 275–283.

Ravet, J. (2018). "But how do I teach them?" Autism & Initial Teacher Education (ITE). *International Journal of Inclusive Education*, *22*(7), 714–733.

Reyes, M. R., Brackett, M. A., Rivers, S. E., Elbertson, N. A., & Salovey, P. (2012). The interaction effects of program training, dosage, and implementation quality on targeted student outcomes for the RULER approach to social and emotional learning. *School Psychology Review*, *41*(1), 82–99.

Ritter, A. B., Terjesen, M. D., & Khuc, T. N. (2019). Autism in Vietnam: Knowledge, training, and communication barriers among professionals, educators, and caregivers. *International Journal of School & Educational Psychology*, *7*(2), 123–134.

Sørensen, K., & Pleasant, A. (2017). Understanding the conceptual importance of the differences among health literacy definitions. Studies *in Health Technology and Informatics*, *240*, 3–14.

Sein, M. K., Henfridsson, O., Purao, S., Rossi, M., & Lindgren, R. (2011). Action design research. *MIS Quarterly*, *35*(1), 37–56.

Shoulders, T. L., & Krei, M. S. (2016). Rural secondary educators' perceptions of their efficacy in the inclusive classroom. *Rural Special Education Quarterly*, *35*(1), 23–30.

Spikol, A., McAteer, D., & Murphy, J. (2019). Recognising autism: A latent transition analysis of parental reports of child autistic spectrum disorder 'red flag' traits before and after age 3. *Social Psychiatry and Psychiatric Epidemiology*, *54*(6), 703–713.

Squires, A., Aiken, L. H., Van den Heede, K., Sermeus, W., Bruyneel, L., Lindqvist, R., . . . Ensio, A. (2013). A systematic survey instrument translation process for multi-country, comparative health workforce studies. *International Journal of Nursing Studies*, *50*(2), 264–273.

Swartz, T., Jehan, F., Tang, A., Gries, L., Zeeshan, M., Kulvatunyou, N., . . . Joseph, B. (2018). Prospective evaluation of low health literacy and its impact on outcomes in trauma patients. *Journal of Trauma and Acute Care Surgery*, *85*(1), 187–192.

Talahun, D., Fekadu, A., Tekola, B., Araya, M., Roth, I., Davey, B., Hanlon, C., & Hoekstra, R. A. (2019). Ethiopian community health workers' beliefs and attitudes towards children with autism: Impact of a brief training intervention. *Autism*, *23*(1), 39–49.

Thunberg, G., Ferm, U., Blom, Å., Karlsson, M., & Nilsson, S. (2019). Implementation of pictorial support for communication with people who have been forced to flee: Experiences from neonatal care. *Journal of Child Health Care*, *23*(2), 311–336.

van Weert, J. C., van Noort, G., Bol, N., van Dijk, L., Tates, K., & Jansen, J. (2011). Tailored information for cancer patients on the Internet: effects of visual cues and language complexity on information recall and satisfaction. *Patient Education and Counseling*, *84*(3), 368–378.

Weaver, S. B., Maneno, M., Chijioke, P., Barsh, D., & Mayaka, R. (2019). Factors associated with health literacy and medication adherence in an urban community. *The Senior Care Pharmacist*, *34*(9), 604–612.

Yu, L., Stronach, S., & Harrison, A. J. (2020). Public knowledge and stigma of autism spectrum disorder: Comparing China with the United States. *Autism*, *24*(6), 1531–1545.